SCIENCE ALL SETS 2024

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CHAPTER: 1 - Chemical Reactions and Equations (Chemistry)
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YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: $Zn + 2CH_3COOH \rightarrow (CH_3COO)_2Zn + H_2$ is a:
OPTIONS: (A) Decomposition reaction (B) Displacement reaction (C) Double displacement reaction (D) Combination reaction
ANSWER: (B) Displacement reaction

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: An aqueous solution of a salt turns blue litmus to red. The salt could be the one obtained by the reaction of:
OPTIONS: (A) HNO_3 and NaOH (B) H_2SO_4 and KOH (C) CH_3COOH and NaOH (D) HCl and NH_4OH
ANSWER: (D) HCl and NH_4OH

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Four solutions, namely glucose, alcohol, hydrochloric acid and sulphuric acid filled in four separate beakers are connected one by one in an electric circuit with a bulb. The solutions in which the bulb will glow when current is passed are:
OPTIONS: (A) Glucose and alcohol (B) Alcohol and hydrochloric acid (C) Glucose and sulphuric acid (D) Hydrochloric acid and sulphuric acid
ANSWER: (D) Hydrochloric acid and sulphuric acid

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Select from the following a decomposition reaction in which source of energy for decomposition is light:
OPTIONS: (A) $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (B) $2H_2O \rightarrow 2H_2 + O_2$ (C) $2AgBr \rightarrow 2Ag + Br_2$ (D) $CaCO_3 \rightarrow CaO + CO_2$
ANSWER: (C) $2AgBr \rightarrow 2Ag + Br_2$

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: When 2 mL of sodium hydroxide solution is added to few pieces of granulated zinc in a test tube and then warmed, the reaction that occurs can be written in the form of a balanced chemical equation as:
OPTIONS: (A) $NaOH + Zn \rightarrow NaZnO_2 + H_2O$ (B) $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$ (C) $2NaOH + Zn \rightarrow NaZnO_2 + H_2$ (D) $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2O$
ANSWER: (B) $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Consider the following chemical equation: a Al_2O_3 + b $HCl \rightarrow$ c $AlCl_3$ + d H_2O. In order to balance this chemical equation, the values of a, b, c and d must be:
OPTIONS: (A) 1, 6, 2 and 3 (B) 1, 6, 3 and 2 (C) 2, 6, 2 and 3 (D) 2, 6, 3 and 2
ANSWER: (A) 1, 6, 2 and 3

YEAR: 2024

CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which one of the following reactions is different from the remaining three?
OPTIONS: (A) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$ (B) $CaO + H_2O \rightarrow Ca(OH)_2$ (C) $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$ (D) $ZnCl_2 + H_2S \rightarrow ZnS + 2HCl$
ANSWER: (B) $CaO + H_2O \rightarrow Ca(OH)_2$

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Identify the product 'X' obtained in the following chemical reaction: $CaCO_3 \rightarrow X + CO_2$
OPTIONS: (A) Quick lime (B) Gypsum (C) Lime Stone (D) Plaster of Paris
ANSWER: (A) Quick lime

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a redox reaction, but not a combination reaction?
OPTIONS: (A) $C + O_2 \rightarrow CO_2$ (B) $2H_2 + O_2 \rightarrow 2H_2O$ (C) $2Mg + O_2 \rightarrow 2MgO$ (D) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
ANSWER: (D) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: To balance the following chemical equation, the values of the coefficients x, y and z must be respectively: $x\ Zn(NO_3)_2 \rightarrow y\ ZnO + z\ NO_2 + O_2$
OPTIONS: (A) 4, 2, 2 (B) 4, 4, 2 (C) 2, 2, 4 (D) 2, 4, 2
ANSWER: (C) 2, 2, 4

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Identify the correct statement about the following reaction: $2H_2S + SO_2 \rightarrow 2H_2O + S$
OPTIONS: (A) H_2S is oxidising agent and SO_2 is reducing agent (B) H_2S is reduced to sulphur (C) SO_2 is oxidising agent and H_2S is reducing agent (D) SO_2 is oxidised to sulphur
ANSWER: (C) SO_2 is oxidising agent and H_2S is reducing agent

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YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The reaction $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$ is a redox reaction because in this case:
OPTIONS: (A) MnO_2 is oxidised and HCl is reduced (B) HCl is oxidised (C) MnO_2 is reduced (D) MnO_2 is reduced and HCl is oxidised
ANSWER: (D) MnO_2 is reduced and HCl is oxidised

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The products obtained when Lead nitrate is heated in a boiling tube are:
OPTIONS: (A) PbO, N_2O and O_2 (B) NO, PbO and O_2 (C) $Pb(NO_2)_2$ and O_2 (D) NO_2, PbO and O_2
ANSWER: (D) NO_2, PbO and O_2

YEAR: 2024

CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is not a thermal decomposition reaction?
OPTIONS: (A) $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (B) $CaCO_3 \rightarrow CaO + CO_2$ (C) $2AgCl \rightarrow 2Ag + Cl_2$ (D) $Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
ANSWER: (C) $2AgCl \rightarrow 2Ag + Cl_2$

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Select from the following a process in which a combination reaction is involved:
OPTIONS: (A) Black and White photography (B) Burning of coal (C) Burning of methane (D) Digestion of food
ANSWER: (B) Burning of coal

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a redox reaction?
OPTIONS: (A) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$ (B) $CaO + H_2O \rightarrow Ca(OH)_2$ (C) $2H_2 + O_2 \rightarrow 2H_2O$ (D) $NaOH + HCl \rightarrow NaCl + H_2O$
ANSWER: (C) $2H_2 + O_2 \rightarrow 2H_2O$

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Name the type of chemical reaction in which calcium oxide reacts with water. Justify your answer by giving balanced chemical equation for the chemical reaction.

OPTIONS: (A) Combination reaction, $CaO + H_2O \rightarrow Ca(OH)_2$ + Heat (B) Decomposition reaction, $CaO + H_2O \rightarrow Ca(OH)_2$ + Heat (C) Displacement reaction, $CaO + H_2O \rightarrow Ca(OH)_2$ (D) Double displacement reaction, $CaO + H_2O \rightarrow Ca(OH)_2$
ANSWER: (A) Combination reaction, $CaO + H_2O \rightarrow Ca(OH)_2$ + Heat

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YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Translate the following statement into a balanced chemical equation: When barium chloride reacts with aluminium sulphate, aluminium chloride and barium sulphate are formed. State the type of this reaction.
OPTIONS: (A) $BaCl_2 + Al_2(SO_4)_3 \rightarrow AlCl_3 + BaSO_4$, Displacement reaction (B) $3BaCl_2 + Al_2(SO_4)_3 \rightarrow 2AlCl_3 + 3BaSO_4$, Precipitation reaction (C) $3BaCl_2 + Al_2(SO_4)_3 \rightarrow 2AlCl_3 + 3BaSO_4$, Double displacement reaction (D) $BaCl_2 + Al_2(SO_4)_3 \rightarrow AlCl_3 + BaSO_4$, Combination reaction
ANSWER: (C) $3BaCl_2 + Al_2(SO_4)_3 \rightarrow 2AlCl_3 + 3BaSO_4$, Double displacement reaction

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YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Translate the following statements into chemical equations and then balance them: (i) Hydrogen sulphide gas burns in air to give water and sulphur dioxide. (ii) Silver bromide on exposure to sunlight decomposes into silver and bromine.
OPTIONS: (A) $2H_2S + 3O_2 \rightarrow 2SO_2 + 2H_2O$; $2AgBr \rightarrow 2Ag + Br_2$ (B) $H_2S + O_2 \rightarrow SO_2 + H_2O$; $AgBr \rightarrow Ag + Br_2$ (C) $2H_2S + O_2 \rightarrow SO_2 + H_2O$; $2AgBr \rightarrow 2Ag + Br_2$ (D) $H_2S + 3O_2 \rightarrow 2SO_2 + H_2O$; $AgBr \rightarrow Ag + Br_2$
ANSWER: (A) $2H_2S + 3O_2 \rightarrow 2SO_2 + 2H_2O$; $2AgBr \rightarrow 2Ag + Br_2$

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2

QUESTION: A spatula full of sodium carbonate is taken in a test tube and 2 mL of dilute ethanoic acid is added to it. Write a chemical equation for the reaction and suggest a method of testing the gas liberated in the reaction.
OPTIONS: (A) $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$; Pass the gas in lime water (B) $CH_3COOH + Na_2CO_3 \rightarrow CH_3COONa + H_2O + CO_2$; Bring a burning matchstick near the gas (C) $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$; Bring a burning matchstick near the gas (D) $CH_3COOH + Na_2CO_3 \rightarrow CH_3COONa + H_2O + CO_2$; Pass the gas in lime water
ANSWER: (A) $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$; Pass the gas in lime water

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YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Write one chemical equation each for the chemical reaction in which the following have taken place: (i) Change in colour (ii) Change in temperature (iii) Formation of precipitate. Mention colour change/temperature change/compound precipitated along with equation.
OPTIONS: (A) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Blue to green); $NaOH + HCl \rightarrow NaCl + H_2O$ + Heat; $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$ (Yellow ppt) (B) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Green to blue); $NaOH + HCl \rightarrow NaCl + H_2O$ + Heat; $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$ (White ppt) (C) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Blue to green); $NaOH + HCl \rightarrow NaCl + H_2O$ + Heat; $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$ (Yellow ppt) (D) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Blue to green); $NaOH + HCl \rightarrow NaCl + H_2O$ - Heat; $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$ (Yellow ppt)
ANSWER: (A) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Blue to green); $NaOH + HCl \rightarrow NaCl + H_2O$ + Heat; $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$ (Yellow ppt)

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: It is observed that Calcium on reaction with water floats on its surface. Explain why it happens. Also write a balanced chemical equation for the reaction that

occurs. What happens when the aqueous solution of the product of this reaction reacts with Carbon dioxide gas? Write a balanced chemical equation for the reaction.
OPTIONS: (A) Bubbles of hydrogen gas stick to calcium and make it lighter; $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$; Solution turns milky; $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ (B) Bubbles of oxygen gas stick to calcium and make it lighter; $Ca + H_2O \rightarrow CaO + H_2$; Solution turns milky; $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ (C) Bubbles of hydrogen gas stick to calcium and make it lighter; $Ca + H_2O \rightarrow Ca(OH)_2 + H_2$; Solution turns milky; $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ (D) Bubbles of hydrogen gas stick to calcium and make it lighter; $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$; Solution turns milky; $CaO + CO_2 \rightarrow CaCO_3$
ANSWER: (A) Bubbles of hydrogen gas stick to calcium and make it lighter; $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$; Solution turns milky; $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: When lead nitrate is heated strongly in a boiling tube, two gases are liberated and a solid residue is left behind in the test tube. (i) Name the type of chemical reaction and define it. (ii) Write the name and formula of the coloured gas liberated. (iii) Write the balanced chemical equation for the reaction. (iv) Name the residue left in the test tube and state the method of testing its nature (acidic/basic).
OPTIONS: (A) Decomposition reaction (single reactant breaks down); Nitrogen dioxide, NO_2; $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$; Lead oxide, dissolve in water and test with litmus (turns blue - basic) (B) Displacement reaction; Nitrogen dioxide, NO_2; $Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$; Lead oxide, dissolve in water and test with litmus (turns red - acidic) (C) Decomposition reaction; Nitrogen dioxide, NO_2; $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$; Lead oxide, dissolve in water and test with litmus (turns red - acidic) (D) Combination reaction; Nitrogen dioxide, NO_2; $Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$; Lead oxide, dissolve in water and test with litmus (turns blue - basic)
ANSWER: (A) Decomposition reaction (single reactant breaks down); Nitrogen dioxide, NO_2; $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$; Lead oxide, dissolve in water and test with litmus (turns blue - basic)

YEAR: 2024
CHAPTER: 1 - Chemical Reactions and Equations
ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5
QUESTION: (i) Write balanced chemical equation for the following word equation: Lead nitrate + Potassium iodide → Lead iodide + Potassium nitrate. Is this a double displacement reaction? Justify your answer. Name the compound precipitated and write the ions present in it. (ii) Write the method of preparation of Ca(OH)2. What happens when CO2 is passed through it? Write balanced chemical equation for the reaction involved.
OPTIONS: (A) Pb(NO3)2 + 2KI → PbI2 + 2KNO3; Yes, exchange of ions takes place; Lead iodide, Pb2+ and I-; CaO + H2O → Ca(OH)2; Turns milky; Ca(OH)2 + CO2 → CaCO3 + H2O (B) Pb(NO3)2 + 2KI → PbI2 + 2KNO3; No, exchange of ions does not take place; Lead iodide, Pb2+ and I-; CaO + H2O → Ca(OH)2; Turns milky; Ca(OH)2 + CO2 → CaCO3 + H2O (C) Pb(NO3)2 + 2KI → PbI2 + 2KNO3; Yes, exchange of ions takes place; Lead iodide, Pb2+ and I-; Ca + 2H2O → Ca(OH)2 + H2; Turns milky; Ca(OH)2 + CO2 → CaCO3 + H2O (D) Pb(NO3)2 + KI → PbI + KNO3; Yes, exchange of ions takes place; Lead iodide, Pb2+ and I-; CaO + H2O → Ca(OH)2; Turns milky; Ca(OH)2 + CO2 → CaCO3 + H2O
ANSWER: (A) Pb(NO3)2 + 2KI → PbI2 + 2KNO3; Yes, exchange of ions takes place; Lead iodide, Pb2+ and I-; CaO + H2O → Ca(OH)2; Turns milky; Ca(OH)2 + CO2 → CaCO3 + H2O

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CHAPTER: 2 - Acids, Bases and Salts (Chemistry)
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YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The salt present in tooth enamel is:
OPTIONS: (A) Calcium phosphate (B) Magnesium phosphate (C) Sodium phosphate (D) Aluminium phosphate
ANSWER: (A) Calcium phosphate

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: A chemical compound used in glass, soap and paper industries is:
OPTIONS: (A) Washing Soda (B) Baking Soda (C) Bleaching Powder (D) Common Salt
ANSWER: (A) Washing Soda

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Solid Calcium oxide reacts vigorously with water to form Calcium hydroxide accompanied by the liberation of heat. From the information given above it may be concluded that this reaction is:
OPTIONS: (A) endothermic and pH of the solution formed is more than 7 (B) exothermic and pH of the solution formed is 7 (C) endothermic and pH of the solution formed is 7 (D) exothermic and pH of the solution formed is more than 7
ANSWER: (D) exothermic and pH of the solution formed is more than 7

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Juice of tamarind turns blue litmus to red. It is because of the presence of an acid called:
OPTIONS: (A) methanoic acid (B) acetic acid (C) tartaric acid (D) oxalic acid
ANSWER: (C) tartaric acid

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: An aqueous solution of sodium chloride is prepared in distilled water. The pH of this solution is:
OPTIONS: (A) 6 (B) 8 (C) 7 (D) 3
ANSWER: (C) 7

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The oxide which can react with HCl as well as KOH to give corresponding salt and water is:
OPTIONS: (A) CuO (B) Al2O3 (C) Na2O (D) K2O
ANSWER: (B) Al2O3

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Tooth decay begins at the pH of:
OPTIONS: (A) 5.1 (B) 5.8 (C) 6.5 (D) 8.0
ANSWER: (A) 5.1

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following salts does not contain water of crystallisation?
OPTIONS: (A) Washing soda (B) Baking soda (C) Gypsum (D) Plaster of Paris
ANSWER: (B) Baking soda

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The pH of a solution is 2. What would be its nature?
OPTIONS: (A) Strongly acidic (B) Strongly basic (C) Weakly acidic (D) Neutral

ANSWER: (A) Strongly acidic

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: The pH of a sample of tomato juice is 4.6. How is this juice likely to be in taste? Give reason. How do we differentiate between a strong acid and a weak base in terms of ion-formation in aqueous solutions? The acid rain can make the survival of aquatic animals difficult. How?
OPTIONS: (A) Slightly sour; acids are sour in taste; Strong acids give more H+ ions, weak bases give less OH- ions; Aquatic animals cannot survive below pH 5.6 (B) Slightly sweet; acids are sweet in taste; Strong acids give more H+ ions, weak bases give less OH- ions; Aquatic animals cannot survive below pH 5.6 (C) Slightly sour; acids are sour in taste; Strong acids give less H+ ions, weak bases give more OH- ions; Aquatic animals cannot survive below pH 5.6 (D) Slightly sour; acids are sour in taste; Strong acids give more H+ ions, weak bases give less OH- ions; Aquatic animals cannot survive below pH 7.0
ANSWER: (A) Slightly sour; acids are sour in taste; Strong acids give more H+ ions, weak bases give less OH- ions; Aquatic animals cannot survive below pH 5.6

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: Salts play a very important role in our daily life. Sodium chloride which is known as common salt is used almost in every kitchen. Baking soda is also a salt used in faster cooking as well as in baking industry. The family of salts is classified on the basis of cations and anions present in them. (a) Identify the acid and base from which Sodium chloride is formed. (b) Find the cation and the anion present in Calcium sulphate. (c) "Sodium chloride and washing soda both belong to the same family of salts." Justify this statement.
OPTIONS: (A) (a) HCl and NaOH; (b) Ca2+ and SO42-; (c) Both have same cation Na+ (B) (a) H2SO4 and NaOH; (b) Ca2+ and SO42-; (c) Both have same anion Cl- (C) (a) HCl and NaOH; (b) Ca2+ and SO42-; (c) Both have same anion SO42- (D) (a) CH3COOH and NaOH; (b) Ca2+ and SO42-; (c) Both have same cation Na+

ANSWER: (A) (a) HCl and NaOH; (b) Ca^{2+} and SO_4^{2-}; (c) Both have same cation Na^+

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: Define the term pH scale. Name the salt obtained by the reaction of Potassium hydroxide and Sulphuric acid and give the pH value of its aqueous solution.
OPTIONS: (A) Scale for measuring H^+ ion concentration; Potassium sulphate, K_2SO_4; pH = 7 (B) Scale for measuring OH^- ion concentration; Potassium sulphate, K_2SO_4; pH = 7 (C) Scale for measuring H^+ ion concentration; Potassium chloride, KCl; pH = 7 (D) Scale for measuring H^+ ion concentration; Potassium sulphate, K_2SO_4; pH = 5
ANSWER: (A) Scale for measuring H^+ ion concentration; Potassium sulphate, K_2SO_4; pH = 7

YEAR: 2024
CHAPTER: 2 - Acids, Bases and Salts
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: What is the chemical name of baking soda? How is it prepared? Write the chemical equation. What happens when baking soda is heated? Give the chemical equation.
OPTIONS: (A) Sodium hydrogen carbonate; $NaCl + H_2O + CO_2 + NH_3 \rightarrow NaHCO_3 + NH_4Cl$; $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ (B) Sodium carbonate; $NaCl + H_2O + CO_2 + NH_3 \rightarrow Na_2CO_3 + NH_4Cl$; $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ (C) Sodium hydrogen carbonate; $NaCl + H_2O + CO_2 + NH_3 \rightarrow NaHCO_3 + NH_4Cl$; $NaHCO_3 \rightarrow NaOH + CO_2$ (D) Sodium hydrogen carbonate; $NaCl + H_2O + CO_2 + NH_3 \rightarrow NaHCO_3 + NH_4Cl$; $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$
ANSWER: (A) Sodium hydrogen carbonate; $NaCl + H_2O + CO_2 + NH_3 \rightarrow NaHCO_3 + NH_4Cl$; $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

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CHAPTER: 3 - Metals and Non-metals (Chemistry)
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YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A metal and a non-metal that exists in liquid state at the room temperature are respectively:
OPTIONS: (A) Bromine and Mercury (B) Mercury and Iodine (C) Mercury and Bromine (D) Iodine and Mercury
ANSWER: (C) Mercury and Bromine

YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A metal X is used in thermit process. When X is heated with oxygen, it gives an oxide Y, which is amphoteric in nature. X and Y respectively are:
OPTIONS: (A) Mn, $MnO2$ (B) Al, $Al2O3$ (C) Fe, $Fe2O3$ (D) Mg, MgO
ANSWER: (B) Al, $Al2O3$

YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Oxides of aluminium and zinc are:
OPTIONS: (A) acidic (B) basic (C) amphoteric (D) neutral
ANSWER: (C) amphoteric

YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is an alloy of copper and tin?
OPTIONS: (A) Nichrome (B) Brass (C) Constantan (D) Bronze

ANSWER: (D) Bronze

YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The metals which are found in both free state as well as combined state are:
OPTIONS: (A) Gold and platinum (B) Platinum and silver (C) Copper and silver (D) Gold and silver
ANSWER: (C) Copper and silver

YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following metals is most reactive?
OPTIONS: (A) Iron (B) Copper (C) Sodium (D) Silver
ANSWER: (C) Sodium

YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is the best conductor of electricity?
OPTIONS: (A) Iron (B) Copper (C) Silver (D) Aluminium
ANSWER: (C) Silver

YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3

QUESTION: State reasons for the following: (a) Zinc oxide is an amphoteric oxide. (b) Sodium metal is stored in bottle filled with kerosene oil. (c) In the reactions of nitric acid with metals, generally hydrogen gas is not evolved.
OPTIONS: (A) (a) Reacts with acids and bases (b) Reacts vigorously with air (c) Nitric acid is a strong oxidising agent; oxidises H2 to water (B) (a) Reacts with acids only (b) Reacts vigorously with air (c) Nitric acid is a strong oxidising agent; oxidises H2 to water (C) (a) Reacts with acids and bases (b) Reacts vigorously with air (c) Nitric acid is a weak oxidising agent; does not oxidise H2 (D) (a) Reacts with bases only (b) Reacts vigorously with air (c) Nitric acid is a strong oxidising agent; oxidises H2 to water
ANSWER: (A) (a) Reacts with acids and bases (b) Reacts vigorously with air (c) Nitric acid is a strong oxidising agent; oxidises H2 to water

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YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: State giving reason the reduction process to obtain the following metals from their compounds: (i) Mercury, (ii) Copper and (iii) Sodium.
OPTIONS: (A) (i) Roasting - low reactivity (ii) Roasting - low reactivity (iii) Electrolytic reduction - high reactivity (B) (i) Roasting - low reactivity (ii) Roasting - low reactivity (iii) Roasting - high reactivity (C) (i) Electrolytic reduction - low reactivity (ii) Roasting - low reactivity (iii) Electrolytic reduction - high reactivity (D) (i) Roasting - low reactivity (ii) Electrolytic reduction - low reactivity (iii) Electrolytic reduction - high reactivity
ANSWER: (A) (i) Roasting - low reactivity (ii) Roasting - low reactivity (iii) Electrolytic reduction - high reactivity

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YEAR: 2024
CHAPTER: 3 - Metals and Non-metals
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: The metals produced by various reduction processes are not very pure. They contain impurities, which must be removed to obtain pure metals. The most widely used method for refining impure metals is electrolytic refining. (i) What is the cathode and anode made of in the refining of copper by this process? (ii) Name the solution used in the above process and write its formula. (iii) How copper gets refined when electric current is passed in the electrolytic cell?

OPTIONS: (A) (i) Cathode - pure copper, Anode - impure copper; (ii) Acidified copper sulphate, $CuSO_4$; (iii) Pure copper dissolves from anode and deposited on cathode (B) (i) Cathode - impure copper, Anode - pure copper; (ii) Acidified copper sulphate, $CuSO_4$; (iii) Pure copper dissolves from anode and deposited on cathode (C) (i) Cathode - pure copper, Anode - impure copper; (ii) Acidified copper sulphate, $CuSO_4$; (iii) Impure copper dissolves from anode and deposited on cathode (D) (i) Cathode - pure copper, Anode - impure copper; (ii) Acidified copper chloride, $CuCl_2$; (iii) Pure copper dissolves from anode and deposited on cathode
ANSWER: (A) (i) Cathode - pure copper, Anode - impure copper; (ii) Acidified copper sulphate, $CuSO_4$; (iii) Pure copper dissolves from anode and deposited on cathode

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CHAPTER: 4 - Carbon and its Compounds (Chemistry)
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YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Carbon compounds: (i) are good conductors of electricity. (ii) are bad conductors of electricity. (iii) have strong forces of attraction between their molecules. (iv) have weak forces of attraction between their molecules. The correct statements are:
OPTIONS: (A) (i) and (ii) (B) (ii) and (iii) (C) (ii) and (iv) (D) (i) and (iii)
ANSWER: (C) (ii) and (iv)

YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Consider the following statements about homologous series of carbon compounds: (a) All succeeding members differ by $-CH_2$ unit. (b) Melting point and boiling point increases with increasing molecular mass. (c) The difference in molecular masses between two successive members is 16 u. (d) C_2H_2 and C_3H_4 are NOT the successive members of alkyne series. The correct statements are:
OPTIONS: (A) (a) and (b) (B) (b) and (c) (C) (a) and (c) (D) (c) and (d)
ANSWER: (A) (a) and (b)

YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The name and formula of third member of homologous series of alkyne is:
OPTIONS: (A) Propyne C_3H_6 (B) Propyne C_3H_4 (C) Butyne C_4H_8 (D) Butyne C_4H_6
ANSWER: (D) Butyne C_4H_6

YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The number of single and double bonds present in a molecule of benzene (C_6H_6) respectively, are:
OPTIONS: (A) 6 and 6 (B) 9 and 3 (C) 3 and 9 (D) 3 and 3
ANSWER: (B) 9 and 3

YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which one of the following hydrocarbons is different from the others?
OPTIONS: (A) C_4H_{10} (B) C_7H_{14} (C) C_5H_{12} (D) C_2H_6
ANSWER: (B) C_7H_{14}

YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Identify a group of the unsaturated hydrocarbons from the following:

OPTIONS: (A) Propane, Ethene, Butyne (B) Ethene, Propane, Hexane (C) Cyclohexane, Methane, Ethane (D) Butyne, Ethene, Propyne
ANSWER: (D) Butyne, Ethene, Propyne

YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is the molecular formula of ethanoic acid?
OPTIONS: (A) $C2H5OH$ (B) $CH3COOH$ (C) $C2H6$ (D) $CH3OH$
ANSWER: (B) $CH3COOH$

YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Distinguish between a saturated and an unsaturated hydrocarbon by flame test. List the products of combustion reaction of a saturated hydrocarbon.
OPTIONS: (A) Saturated burns with clean blue flame; Unsaturated burns with yellow sooty flame; Products are $CO2$ and $H2O$ (B) Saturated burns with yellow sooty flame; Unsaturated burns with clean blue flame; Products are $CO2$ and $H2O$ (C) Saturated burns with clean blue flame; Unsaturated burns with yellow sooty flame; Products are CO and $H2O$ (D) Saturated burns with clean blue flame; Unsaturated burns with yellow sooty flame; Products are $CO2$ and $H2O$
ANSWER: (A) Saturated burns with clean blue flame; Unsaturated burns with yellow sooty flame; Products are $CO2$ and $H2O$

YEAR: 2024
CHAPTER: 4 - Carbon and its Compounds
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Write chemical equations for the following reactions, giving the conditions for the reaction in each case: (a) Reaction of ethanol with ethanoic acid (b) Reaction of an ester with a base (NaOH) (c) Formation of ethene from ethanol

OPTIONS: (A) (a) $CH_3COOH + C_2H_5OH \xrightarrow{\text{Acid Catalyst}} CH_3COOC_2H_5 + H_2O$; (b) $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$; (c) $C_2H_5OH \xrightarrow{\text{Hot Conc. } H_2SO_4, 443K} CH_2=CH_2 + H_2O$ (B) (a) $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$; (b) $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$; (c) $C_2H_5OH \rightarrow CH_2=CH_2 + H_2O$ (C) (a) $CH_3COOH + C_2H_5OH \xrightarrow{\text{Acid Catalyst}} CH_3COOC_2H_5 + H_2O$; (b) $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$; (c) $C_2H_5OH \xrightarrow{\text{Hot Conc. } H_2SO_4} CH_2=CH_2 + H_2O$ (D) (a) $CH_3COOH + C_2H_5OH \xrightarrow{\text{Acid Catalyst}} CH_3COOC_2H_5 + H_2O$; (b) $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$; (c) $C_2H_5OH \xrightarrow{\text{Hot Conc. } H_2SO_4, 443K} CH_2=CH_2 + H_2O$
ANSWER: (A) (a) $CH_3COOH + C_2H_5OH \xrightarrow{\text{Acid Catalyst}} CH_3COOC_2H_5 + H_2O$; (b) $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$; (c) $C_2H_5OH \xrightarrow{\text{Hot Conc. } H_2SO_4, 443K} CH_2=CH_2 + H_2O$

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CHAPTER: 5 - Life Processes (Biology)
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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The process in which transport of soluble products of photosynthesis takes place in plants is known as:
OPTIONS: (A) Transpiration (B) Evaporation (C) Conduction (D) Translocation
ANSWER: (D) Translocation

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Sense organ in which olfactory receptors are present is:
OPTIONS: (A) Nose (B) Skin (C) Tongue (D) Inner ear
ANSWER: (A) Nose

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The incorrect statement about placenta is:
OPTIONS: (A) It is a disc embedded in the uterine wall (B) It contains villi on the embryo's side of the tissue (C) It has a very small surface area for glucose and oxygen to pass from mother to the embryo (D) The embryo gets nutrition from the mother's blood through it
ANSWER: (C) It has a very small surface area for glucose and oxygen to pass from mother to the embryo

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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The correct sequence of events when someone's hand touches a hot object unconsciously:
OPTIONS: (A) Receptors in skin → Motor neuron → Relay neuron → Sensory neuron → Effector muscle in arm (B) Receptors in skin → Relay neuron → Sensory neuron → Motor neuron → Effector muscle in arm (C) Receptors in skin → Sensory neuron → Relay neuron → Motor neuron → Effector muscle in arm (D) Receptors in skin → Sensory neuron → Effector muscle in arm → Motor neuron → Relay neuron
ANSWER: (C) Receptors in skin → Sensory neuron → Relay neuron → Motor neuron → Effector muscle in arm

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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In human beings, when the process of digestion is completed, the (i) proteins, (ii) carbohydrates, and (iii) fats are respectively finally converted into:
OPTIONS: (A) (i) Amino acids, (ii) glucose and (iii) fatty acids (B) (i) Amino acids, (ii) glucose, (iii) fatty acids and glycerol (C) (i) Glucose, (ii) fatty acids and glycerol, (iii) amino acids (D) (i) Sugars, (ii) amino acids, (iii) fatty acids and glycerol
ANSWER: (B) (i) Amino acids, (ii) glucose, (iii) fatty acids and glycerol

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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Identify an involuntary action from the following:
OPTIONS: (A) Riding a bicycle (B) Picking up a pencil (C) Regular beating of heart (D) Walking in a straight line
ANSWER: (C) Regular beating of heart

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In human respiratory system, when a person breathes in, the position of ribs and diaphragm will be:
OPTIONS: (A) lifted ribs and curve/dome shaped diaphragm (B) lifted ribs and flattened diaphragm (C) relaxed ribs and flattened diaphragm (D) relaxed ribs and curve/dome shaped diaphragm
ANSWER: (B) lifted ribs and flattened diaphragm

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is the function of the human kidney?
OPTIONS: (A) Filtering blood to remove waste products (B) Producing digestive enzymes (C) Pumping blood to different parts of the body (D) Exchanging gases between blood and air
ANSWER: (A) Filtering blood to remove waste products

YEAR: 2024

CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The fluid that is filtered out from the blood in the kidneys is called:
OPTIONS: (A) Urine (B) Lymph (C) Filtrate (D) Plasma
ANSWER: (C) Filtrate

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is the main function of the liver in the human digestive system?
OPTIONS: (A) Produces digestive enzymes (B) Stores bile and releases it into the small intestine (C) Produces bile which helps in digestion of fats (D) Absorbs water from undigested food
ANSWER: (C) Produces bile which helps in digestion of fats

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In which part of the human digestive system does maximum absorption of nutrients take place?
OPTIONS: (A) Stomach (B) Large intestine (C) Small intestine (D) Oesophagus
ANSWER: (C) Small intestine

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The enzyme present in saliva that breaks down starch is called:
OPTIONS: (A) Pepsin (B) Trypsin (C) Amylase (D) Lipase
ANSWER: (C) Amylase

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In plants, water and minerals are transported through:
OPTIONS: (A) Phloem (B) Xylem (C) Cambium (D) Epidermis
ANSWER: (B) Xylem

YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Sometimes while running, the athletes suffer from muscle cramps. Why? How is the respiration in this case different from aerobic respiration?
OPTIONS: (A) Lactic acid formation; Anaerobic respiration occurs due to lack of oxygen; End products are lactic acid + Energy (B) Lactic acid formation; Aerobic respiration occurs due to lack of oxygen; End products are lactic acid + Energy (C) Lactic acid formation; Anaerobic respiration occurs due to lack of oxygen; End products are CO_2 + H_2O + Energy (D) Lactic acid formation; Anaerobic respiration occurs due to lack of oxygen; End products are lactic acid + Energy
ANSWER: (A) Lactic acid formation; Anaerobic respiration occurs due to lack of oxygen; End products are lactic acid + Energy

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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Write the other name given to lymph. State its two functions.
OPTIONS: (A) Tissue fluid/Extracellular fluid; Carries digested fats from intestine; Drains excess fluid from extracellular space back into blood (B) Blood plasma; Carries digested fats from intestine; Drains excess fluid from extracellular space back into blood (C) Tissue fluid/Extracellular fluid; Carries digested fats from intestine; Transports

oxygen (D) Tissue fluid/Extracellular fluid; Carries digested fats from intestine; Drains excess fluid from intracellular space back into blood
ANSWER: (A) Tissue fluid/Extracellular fluid; Carries digested fats from intestine; Drains excess fluid from extracellular space back into blood

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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: We need to water the soil in plants on a regular basis. But it ultimately reaches the leaves of the plant. Explain how this takes place.
OPTIONS: (A) Through transpiration pull and xylem transport; Water lost through stomata creates suction pull; Water is pulled up through xylem (B) Through diffusion and phloem transport; Water lost through stomata creates suction pull; Water is pulled up through phloem (C) Through transpiration pull and phloem transport; Water lost through stomata creates suction pull; Water is pulled up through phloem (D) Through diffusion and xylem transport; Water lost through stomata creates suction pull; Water is pulled up through xylem
ANSWER: (A) Through transpiration pull and xylem transport; Water lost through stomata creates suction pull; Water is pulled up through xylem

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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Name the type of nutrition exhibited by Amoeba. Explain how food is taken in and digested by this organism.
OPTIONS: (A) Heterotrophic/Holozoic; Uses pseudopodia to engulf food forming food vacuole; Complex substances broken down into simpler substances (B) Autotrophic; Uses pseudopodia to engulf food forming food vacuole; Complex substances broken down into simpler substances (C) Heterotrophic/Holozoic; Uses flagella to engulf food forming food vacuole; Complex substances broken down into simpler substances (D) Heterotrophic/Holozoic; Uses pseudopodia to engulf food forming food vacuole; Simple substances broken down into complex substances
ANSWER: (A) Heterotrophic/Holozoic; Uses pseudopodia to engulf food forming food vacuole; Complex substances broken down into simpler substances

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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Give reasons for the following: (i) Alveoli in lungs are richly supplied with blood capillaries. (ii) Respiratory pigment in the blood takes up oxygen and not carbon dioxide. (iii) During anaerobic respiration, a 3-carbon molecule is formed as an end product instead of CO_2 in human beings.
OPTIONS: (A) (i) For efficient exchange of gases (ii) High affinity for oxygen (iii) Lack of oxygen oxidises glucose completely (B) (i) For efficient exchange of gases (ii) High affinity for oxygen (iii) Lack of oxygen does not oxidise glucose completely and forms lactic acid (C) (i) For efficient exchange of gases (ii) High affinity for carbon dioxide (iii) Lack of oxygen does not oxidise glucose completely and forms lactic acid (D) (i) For efficient exchange of gases (ii) High affinity for oxygen (iii) Lack of oxygen does not oxidise glucose completely and forms ethanol
ANSWER: (B) (i) For efficient exchange of gases (ii) High affinity for oxygen (iii) Lack of oxygen does not oxidise glucose completely and forms lactic acid

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YEAR: 2024
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Write the main difference between aerobic and anaerobic respiration. State the pathway which is common for both. Write the overall chemical equation of aerobic respiration and mention the site where this process occurs inside the cells.
OPTIONS: (A) Aerobic uses oxygen, Anaerobic does not use oxygen; Glycolysis (Glucose → Pyruvate); $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy; Mitochondria (B) Aerobic uses oxygen, Anaerobic does not use oxygen; Krebs cycle; $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy; Mitochondria (C) Aerobic uses oxygen, Anaerobic does not use oxygen; Glycolysis (Glucose → Pyruvate); $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy; Cytoplasm (D) Aerobic does not use oxygen, Anaerobic uses oxygen; Glycolysis (Glucose → Pyruvate); $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy; Mitochondria
ANSWER: (A) Aerobic uses oxygen, Anaerobic does not use oxygen; Glycolysis (Glucose → Pyruvate); $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy; Mitochondria

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CHAPTER: 6 - Control and Coordination (Biology)
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YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In a nerve cell, the site where the electrical impulse is converted into a chemical signal is known as:
OPTIONS: (A) Axon (B) Dendrites (C) Neuromuscular junction (D) Cell body
ANSWER: (C) Neuromuscular junction

YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A plant growth inhibitor hormone which causes wilting of leaves is called:
OPTIONS: (A) Auxin (B) Cytokinin (C) Abscisic acid (D) Gibberellin
ANSWER: (C) Abscisic acid

YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following statement(s) is (are) true about human heart?
OPTIONS: (A) Right atrium receives oxygenated blood from lungs through pulmonary artery (B) Left atrium transfers oxygenated blood to left ventricle which sends it to various parts of the body (C) Right atrium receives deoxygenated blood from different parts of the body through vena cava (D) Left atrium transfers oxygenated blood to aorta which sends it to different parts of the body
ANSWER: (C) (b) and (c)

YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is not a function of the nervous system?
OPTIONS: (A) Coordination of body activities (B) Response to stimuli (C) Digestion of food (D) Transmission of impulses
ANSWER: (C) Digestion of food

YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The part of the brain responsible for thinking, reasoning and memory is:
OPTIONS: (A) Cerebellum (B) Medulla (C) Cerebrum (D) Pons
ANSWER: (C) Cerebrum

YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a plant hormone that promotes cell elongation?
OPTIONS: (A) Cytokinin (B) Gibberellin (C) Auxin (D) Abscisic acid
ANSWER: (C) Auxin

YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The gap between two neurons is called:
OPTIONS: (A) Synapse (B) Axon (C) Dendrite (D) Myelin sheath
ANSWER: (A) Synapse

YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Define reflex action. With the help of a flow chart show the path of a reflex action such as sneezing.
OPTIONS: (A) Sudden/spontaneous action in response to stimulus; Stimulus → Receptors → Sensory neuron → Spinal cord (Relay neuron) → Motor neuron → Effector → Response (B) Voluntary action in response to stimulus; Stimulus → Receptors → Sensory neuron → Brain → Motor neuron → Effector → Response (C) Sudden/spontaneous action in response to stimulus; Stimulus → Receptors → Sensory neuron → Brain → Motor neuron → Effector → Response (D) Sudden/spontaneous action in response to stimulus; Stimulus → Receptors → Sensory neuron → Spinal cord (Relay neuron) → Motor neuron → Effector → Response
ANSWER: (D) Sudden/spontaneous action in response to stimulus; Stimulus → Receptors → Sensory neuron → Spinal cord (Relay neuron) → Motor neuron → Effector → Response

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YEAR: 2024
CHAPTER: 6 - Control and Coordination
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Name and state the rule to determine the direction of a: (i) magnetic field produced around a current carrying straight conductor. (ii) force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.
OPTIONS: (A) (i) Right Hand Thumb Rule - thumb points towards direction of current, fingers wrap in direction of field; (ii) Fleming's Left Hand Rule - thumb points in direction of force (B) (i) Fleming's Left Hand Rule - thumb points towards direction of current, fingers wrap in direction of field; (ii) Right Hand Thumb Rule - thumb points in direction of force (C) (i) Right Hand Thumb Rule - thumb points towards direction of current, fingers wrap in direction of field; (ii) Fleming's Right Hand Rule - thumb points in direction of force (D) (i) Right Hand Thumb Rule - thumb points towards direction of field, fingers wrap in direction of current; (ii) Fleming's Left Hand Rule - thumb points in direction of force

ANSWER: (A) (i) Right Hand Thumb Rule - thumb points towards direction of current, fingers wrap in direction of field; (ii) Fleming's Left Hand Rule - thumb points in direction of force

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CHAPTER: 7 - Reproduction (Biology)
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YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Select from the following the conditions responsible for the rapid spread of bread mould on a slice of bread: (i) Formation of large number of spores (ii) Presence of moisture and nutrients in bread (iii) Low temperature (iv) Presence of hyphae
OPTIONS: (A) (i) and (ii) (B) (ii) and (iv) (C) (ii) and (iii) (D) (iii) and (iv)
ANSWER: (A) (i) and (ii)

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The plants that can be raised by the method of vegetative propagation are:
OPTIONS: (A) Sugarcane, roses, grapes (B) Sugarcane, mustard, potato (C) Banana, orange, mustard (D) Papaya, mustard, potato
ANSWER: (A) Sugarcane, roses, grapes

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The part of seed which is a source of food during germination of seed is:
OPTIONS: (A) Cotyledon (B) Radicle (C) Plumule (D) Embryo
ANSWER: (A) Cotyledon

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In which of the following organisms, multiple fission is a means of asexual reproduction?
OPTIONS: (A) Yeast (B) Leishmania (C) Paramoecium (D) Plasmodium
ANSWER: (D) Plasmodium

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A zygote is formed by the fusion of a male gamete and a female gamete. The number of chromosomes in the zygote of a human is:
OPTIONS: (A) 23 (B) 44 (C) 46 (D) 92
ANSWER: (C) 46

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is an example of asexual reproduction?
OPTIONS: (A) Fertilisation (B) Budding in yeast (C) Seed formation (D) Pollination
ANSWER: (B) Budding in yeast

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The male reproductive part of a flower is called:

OPTIONS: (A) Pistil (B) Stamen (C) Ovary (D) Stigma
ANSWER: (B) Stamen

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The female reproductive part of a flower is called:
OPTIONS: (A) Stamen (B) Anther (C) Pistil (D) Filament
ANSWER: (C) Pistil

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is NOT a method of asexual reproduction?
OPTIONS: (A) Budding (B) Fragmentation (C) Fertilisation (D) Spore formation
ANSWER: (C) Fertilisation

YEAR: 2024
CHAPTER: 7 - Reproduction
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Pollination is an important process in sexual reproduction of plants. It is an essential process that facilitates fertilisation in plants. Pollinating agents can be wind, water, insects and birds. Several changes take place in the flower after the fertilization has taken place. (a) Write the main difference between self-pollination and cross-pollination. (b) Name the part of the flower which attracts insects for pollination. What happens to this part after fertilisation? (c) Define fertilisation. What is the fate of ovules and the ovary in a flower after fertilisation?
OPTIONS: (A) (a) Self-pollination - same flower, Cross-pollination - different flowers; (b) Petals, they dry and fall off; (c) Fusion of male and female gametes; Ovule - seed, Ovary - fruit (B) (a) Self-pollination - different flowers, Cross-pollination - same flower; (b) Petals, they dry and fall off; (c) Fusion of male and female gametes; Ovule - seed,

Ovary - fruit (C) (a) Self-pollination - same flower, Cross-pollination - different flowers; (b) Sepals, they dry and fall off; (c) Fusion of male and female gametes; Ovule - seed, Ovary - fruit (D) (a) Self-pollination - same flower, Cross-pollination - different flowers; (b) Petals, they dry and fall off; (c) Fusion of male and female gametes; Ovule - fruit, Ovary - seed
ANSWER: (A) (a) Self-pollination - same flower, Cross-pollination - different flowers; (b) Petals, they dry and fall off; (c) Fusion of male and female gametes; Ovule - seed, Ovary - fruit

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CHAPTER: 8 - Heredity and Evolution (Biology)
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YEAR: 2024
CHAPTER: 8 - Heredity and Evolution
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Consider the following statements: (i) The sex of a child is determined by what it inherits from the mother. (ii) The sex of a child is determined by what it inherits from the father. (iii) The probability of having a male child is more than that of a female child. (iv) The sex of a child is determined at the time of fertilisation when male and female gametes fuse to form a zygote. The correct statements are:
OPTIONS: (A) (i) and (iii) (B) (ii) and (iv) (C) (iii) and (iv) (D) (i), (iii) and (iv)
ANSWER: (B) (ii) and (iv)

YEAR: 2024
CHAPTER: 8 - Heredity and Evolution
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Chromosomes: (i) carry hereditary information from parents to the next generation. (ii) are thread like structures located inside the nucleus of an animal cell. (iii) always exist in pairs in human reproductive cells. (iv) are involved in the process of cell division. The correct statements are:
OPTIONS: (A) (i) and (ii) (B) (iii) and (iv) (C) (i), (ii) and (iv) (D) (i) and (iv)
ANSWER: (C) (i), (ii) and (iv)

YEAR: 2024
CHAPTER: 8 - Heredity and Evolution
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A cross made between two pea plants produces 50% tall and 50% short pea plants. The gene combination of the parental pea plants must be:
OPTIONS: (A) Tt and Tt (B) TT and Tt (C) Tt and tt (D) TT and tt
ANSWER: (C) Tt and tt

YEAR: 2024
CHAPTER: 8 - Heredity and Evolution
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In an experiment to study independent inheritance of two separate traits: shape and colour of seeds, the ratio of the different combinations in F2 progeny would be:
OPTIONS: (A) 1:3 (B) 1:2:1 (C) 9:3:3:1 (D) 9:1:1:3
ANSWER: (C) 9:3:3:1

YEAR: 2024
CHAPTER: 8 - Heredity and Evolution
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a dominant trait in pea plants?
OPTIONS: (A) Dwarf height (B) Wrinkled seeds (C) Round seeds (D) Green seeds
ANSWER: (C) Round seeds

YEAR: 2024
CHAPTER: 8 - Heredity and Evolution
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a recessive trait in pea plants?

OPTIONS: (A) Tall height (B) Yellow seeds (C) Wrinkled seeds (D) Axial flowers
ANSWER: (C) Wrinkled seeds

YEAR: 2024
CHAPTER: 8 - Heredity and Evolution
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Mendel crossed pure tall pea plants (TT) with pure short pea plants (tt) and obtained F1 progeny. When the plants of F1 progeny were self-pollinated, plants of F2 progeny were obtained. (a) What did the plants of F1 progeny look like? Give their gene combination. (b) Why could the gene for shortness not be expressed in plants of F1 progeny? (c) Write the ratio of the plants obtained in F2 progeny and state the conclusion that can be drawn from this experiment.
OPTIONS: (A) (a) All tall, Tt; (b) It is recessive trait; (c) Tall:Short = 3:1; Conclusion: Tall is dominant, short is recessive (B) (a) All tall, TT; (b) It is recessive trait; (c) Tall:Short = 1:1; Conclusion: Tall is dominant, short is recessive (C) (a) All short, tt; (b) It is recessive trait; (c) Tall:Short = 3:1; Conclusion: Short is dominant, tall is recessive (D) (a) All tall, Tt; (b) It is dominant trait; (c) Tall:Short = 3:1; Conclusion: Tall is dominant, short is recessive
ANSWER: (A) (a) All tall, Tt; (b) It is recessive trait; (c) Tall:Short = 3:1; Conclusion: Tall is dominant, short is recessive

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CHAPTER: 9 - Light - Reflection and Refraction (Physics)
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YEAR: 2024
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: At what distance from a convex lens should an object be placed to get an image of the same size as that of the object on a screen?
OPTIONS: (A) Beyond twice the focal length of the lens (B) At the principal focus of the lens (C) At twice the focal length of the lens (D) Between the optical centre of the lens and its principal focus
ANSWER: (C) At twice the focal length of the lens

YEAR: 2024
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The lens system of human eye forms an image on a light sensitive screen, which is called as:
OPTIONS: (A) Cornea (B) Ciliary muscles (C) Optic nerves (D) Retina
ANSWER: (D) Retina

YEAR: 2024
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: How will the image formed by a convex lens be affected, if the upper half of the lens is wrapped with a black paper?
OPTIONS: (A) The size of the image formed will be one-half of the size of the image due to complete lens (B) The image of upper half of the object will not be formed (C) The brightness of the image will reduce (D) The lower half of the inverted image will not be formed
ANSWER: (C) The brightness of the image will reduce

YEAR: 2024
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Absolute refractive index of glass and water is 3/2 and 4/3 respectively. If the speed of light in glass is 2 x 10^8 m/s, the speed of light in water is:
OPTIONS: (A) 9/4 x 10^8 m/s (B) 5/2 x 10^8 m/s (C) 7/3 x 10^8 m/s (D) 16/9 x 10^8 m/s
ANSWER: (A) 9/4 x 10^8 m/s

YEAR: 2024

CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The focal length of a concave mirror is 20 cm. What is its radius of curvature?
OPTIONS: (A) 10 cm (B) 20 cm (C) 40 cm (D) 60 cm
ANSWER: (C) 40 cm

YEAR: 2024
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A convex lens of focal length 20 cm forms a real image at a distance of 40 cm from the lens. The object distance is:
OPTIONS: (A) 20 cm (B) 40 cm (C) 60 cm (D) 80 cm
ANSWER: (B) 40 cm

YEAR: 2024
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The power of a lens is +2 D. The focal length of the lens is:
OPTIONS: (A) +0.5 m (B) -0.5 m (C) +2 m (D) -2 m
ANSWER: (A) +0.5 m

YEAR: 2024
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: A highly polished surface such as a mirror reflects most of the light falling on it. In our daily life we use two types of mirrors - plane and spherical. The reflecting surface of a spherical mirrors may be curved inwards or outwards. In concave mirrors, reflection takes place from the inner surface, while in convex mirrors reflection takes place from the outer surface. (a) Define the principal axis of a concave mirror. (b) A ray

of light is incident on a concave mirror, parallel to its principal axis. If this ray after reflection from the mirror passes through the principal axis from a point at a distance of 10 cm from the pole of the mirror, find the radius of curvature of the mirror. (c) An object is placed at a distance of 10 cm from the pole of a convex mirror of focal length 15 cm. Find the position of the image.
OPTIONS: (A) (a) Straight line passing through pole and centre of curvature; (b) R = 20 cm; (c) v = +6 cm (behind the mirror) (B) (a) Straight line passing through pole and centre of curvature; (b) R = 20 cm; (c) v = -6 cm (in front of the mirror) (C) (a) Straight line passing through pole and centre of curvature; (b) R = 10 cm; (c) v = +6 cm (behind the mirror) (D) (a) Straight line passing through pole and centre of curvature; (b) R = 20 cm; (c) v = +15 cm (behind the mirror)
ANSWER: (A) (a) Straight line passing through pole and centre of curvature; (b) R = 20 cm; (c) v = +6 cm (behind the mirror)

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CHAPTER: 10 - Human Eye and Colourful World (Physics)
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YEAR: 2024
CHAPTER: 10 - Human Eye and Colourful World
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The phenomena of light involved in the formation of rainbow are:
OPTIONS: (A) Refraction, reflection and dispersion (B) Refraction, dispersion and internal reflection (C) Reflection, dispersion and internal reflection (D) Refraction, dispersion, scattering and total internal reflection
ANSWER: (B) Refraction, dispersion and internal reflection

YEAR: 2024
CHAPTER: 10 - Human Eye and Colourful World
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The colour of light for which the refractive index of glass is minimum, is:
OPTIONS: (A) Red (B) Yellow (C) Green (D) Violet
ANSWER: (A) Red

YEAR: 2024
CHAPTER: 10 - Human Eye and Colourful World
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is:
OPTIONS: (A) Red (B) Orange (C) Blue (D) Yellow
ANSWER: (C) Blue

YEAR: 2024
CHAPTER: 10 - Human Eye and Colourful World
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The defect of vision in which a person cannot see distant objects clearly is called:
OPTIONS: (A) Myopia (B) Hypermetropia (C) Presbyopia (D) Astigmatism
ANSWER: (A) Myopia

YEAR: 2024
CHAPTER: 10 - Human Eye and Colourful World
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The defect of vision in which a person cannot see nearby objects clearly is called:
OPTIONS: (A) Myopia (B) Hypermetropia (C) Presbyopia (D) Astigmatism
ANSWER: (B) Hypermetropia

YEAR: 2024
CHAPTER: 10 - Human Eye and Colourful World
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3

QUESTION: Study the diagram and answer the questions that follow: (i) Name the defect of vision depicted in this diagram stating the part of the eye responsible for this condition. (ii) List two causes of this defect. (iii) Name the type of lens used to correct this defect and state its role in this case.
OPTIONS: (A) (i) Hypermetropia, Ciliary muscles/eye lens; (ii) Focal length too long, Eyeball too small; (iii) Converging/Convex lens, provides additional focussing power (B) (i) Myopia, Ciliary muscles/eye lens; (ii) Focal length too short, Eyeball too large; (iii) Diverging/Concave lens, provides additional focussing power (C) (i) Hypermetropia, Ciliary muscles/eye lens; (ii) Focal length too short, Eyeball too large; (iii) Converging/Convex lens, provides additional focussing power (D) (i) Myopia, Ciliary muscles/eye lens; (ii) Focal length too long, Eyeball too small; (iii) Diverging/Concave lens, provides additional focussing power
ANSWER: (A) (i) Hypermetropia, Ciliary muscles/eye lens; (ii) Focal length too long, Eyeball too small; (iii) Converging/Convex lens, provides additional focussing power

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CHAPTER: 11 - Electricity (Physics)
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YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: An electric iron of resistance 20 Ω draws a current of 5 A. The heat developed in the iron in 30 seconds is:
OPTIONS: (A) 15000 J (B) 6000 J (C) 1500 J (D) 3000 J
ANSWER: (A) 15000 J

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The maximum resistance of a network of five identical resistors of 1/5 Ω each can be:
OPTIONS: (A) 1 Ω (B) 0.5 Ω (C) 0.25 Ω (D) 0.1 Ω
ANSWER: (A) 1 Ω

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: S.I. unit of electrical resistivity is:
OPTIONS: (A) ohm per metre3 (B) ohm per metre2 (C) ohm.metre (D) ohm.metre3
ANSWER: (C) ohm.metre

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The resistance of a conductor depends on:
OPTIONS: (A) Length and area of cross-section (B) Temperature only (C) Material only (D) Length, area of cross-section and material
ANSWER: (D) Length, area of cross-section and material

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If the length of a wire is doubled, its resistance becomes:
OPTIONS: (A) Half (B) Double (C) Four times (D) One-fourth
ANSWER: (B) Double

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The power of an electric appliance is given by:
OPTIONS: (A) P = I^2R (B) P = V^2/R (C) P = VI (D) All of the above

ANSWER: (D) All of the above

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The unit of electric energy is:
OPTIONS: (A) Watt (B) Joule (C) Volt (D) Ampere
ANSWER: (B) Joule

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is NOT a source of electrical energy?
OPTIONS: (A) Cell (B) Battery (C) Thermocouple (D) Resistor
ANSWER: (D) Resistor

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Two wires A and B of same material, having same lengths and diameters 0.2 mm and 0.3 mm respectively, are connected one by one in a circuit. Which one of these two wires will offer more resistance to the flow of current in the circuit? Justify your answer.
OPTIONS: (A) Wire A (thinner wire offers more resistance); $R \propto 1/A$ (B) Wire B (thicker wire offers more resistance); $R \propto A$ (C) Wire A (thinner wire offers less resistance); $R \propto 1/A$ (D) Both offer same resistance; $R \propto 1/A$
ANSWER: (A) Wire A (thinner wire offers more resistance); $R \propto 1/A$

YEAR: 2024

CHAPTER: 11 - Electricity
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: An electric source can supply a charge of 500 coulomb. If the current drawn by a device is 25 mA, find the time in which the electric source will be discharged completely.
OPTIONS: (A) 20000 s (B) 2000 s (C) 200000 s (D) 20 s
ANSWER: (A) 20000 s

YEAR: 2024
CHAPTER: 11 - Electricity
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: (i) The potential difference across the two ends of a circuit component is decreased to one-third of its initial value, while its resistance remains constant. What change will be observed in the current flowing through it? Name and state the law which helps us to answer this question. (ii) Draw a schematic diagram of a circuit consisting of a battery of four 1.5 V cells, a 5 Ω resistor, a 10 Ω resistor and a 15 Ω resistor and a plug key, all connected in series. Now find (I) the electric current passing through the circuit, and (II) potential difference across the 10 Ω resistor when the plug key is closed.
OPTIONS: (A) (i) Current becomes one-third; Ohm's Law ($V \propto I$); (ii) I = 0.2 A, V across 10 Ω = 2 V (B) (i) Current becomes three times; Ohm's Law ($V \propto I$); (ii) I = 0.2 A, V across 10 Ω = 2 V (C) (i) Current becomes one-third; Ohm's Law ($V \propto I$); (ii) I = 0.2 A, V across 10 Ω = 3 V (D) (i) Current becomes one-third; Ohm's Law ($V \propto I$); (ii) I = 0.2 A, V across 10 Ω = 2 V
ANSWER: (A) (i) Current becomes one-third; Ohm's Law ($V \propto I$); (ii) I = 0.2 A, V across 10 Ω = 2 V

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CHAPTER: 12 - Magnetic Effects of Current (Physics)
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YEAR: 2024
CHAPTER: 12 - Magnetic Effects of Current
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: The current carrying device which produces a magnetic field similar to that of a bar magnet is:
OPTIONS: (A) A straight conductor (B) A circular loop (C) A solenoid (D) A circular coil
ANSWER: (C) A solenoid

YEAR: 2024
CHAPTER: 12 - Magnetic Effects of Current
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A rectangular loop ABCD carrying a current I is situated near a straight conductor XY, such that the conductor is parallel to the side AB of the loop and is in the plane of the loop. If a steady current I is established in the conductor as shown, the conductor XY will:
OPTIONS: (A) remain stationary (B) move towards the side AB of the loop (C) move away from the side AB of the loop (D) rotate about its axis
ANSWER: (B) move towards the side AB of the loop

YEAR: 2024
CHAPTER: 12 - Magnetic Effects of Current
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The direction of magnetic field produced around a straight current-carrying conductor is given by:
OPTIONS: (A) Fleming's left hand rule (B) Fleming's right hand rule (C) Right hand thumb rule (D) Maxwell's rule
ANSWER: (C) Right hand thumb rule

YEAR: 2024
CHAPTER: 12 - Magnetic Effects of Current
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The strength of magnetic field at a point near a current-carrying conductor depends on:

OPTIONS: (A) Magnitude of current (B) Distance from the conductor (C) Both magnitude of current and distance (D) Direction of current only
ANSWER: (C) Both magnitude of current and distance

YEAR: 2024
CHAPTER: 12 - Magnetic Effects of Current
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following devices uses the magnetic effect of current?
OPTIONS: (A) Electric bulb (B) Electric iron (C) Electric bell (D) Heater
ANSWER: (C) Electric bell

YEAR: 2024
CHAPTER: 12 - Magnetic Effects of Current
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Two magnetic field lines do not intersect each other. Why? How is a uniform magnetic field in a given region represented? Draw a diagram in support of your answer.
OPTIONS: (A) If they intersect, there would be two directions of magnetic field; Uniform field is represented by equidistant parallel straight lines (B) If they intersect, there would be no direction of magnetic field; Uniform field is represented by equidistant parallel straight lines (C) If they intersect, there would be two directions of magnetic field; Uniform field is represented by concentric circles (D) If they intersect, there would be two directions of magnetic field; Uniform field is represented by equidistant parallel straight lines
ANSWER: (A) If they intersect, there would be two directions of magnetic field; Uniform field is represented by equidistant parallel straight lines

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YEAR: 2024
CHAPTER: 12 - Magnetic Effects of Current
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3

QUESTION: A student fixes a sheet of white paper on a drawing board. He places a bar magnet in the centre of it. He sprinkles some iron filings uniformly around the bar magnet. Then he taps the drawing board gently and observes that the iron filings arrange themselves in a particular pattern. (a) Why do iron filings arrange in a particular pattern? (b) What does the crowding of iron filings at the ends of the magnet indicate? (c) What do the lines, along which the iron filings align, represent? (d) If the student places a cardboard horizontally in a current carrying solenoid and repeats the above activity, in what pattern would the iron filings arrange? State the conclusion drawn about the magnetic field based on the observed pattern of the lines.
OPTIONS: (A) (a) Magnetic field exists around magnet; (b) Maximum strength near poles; (c) Magnetic field lines; (d) Equidistant parallel lines; Field inside solenoid is uniform (B) (a) Magnetic field exists around magnet; (b) Maximum strength near poles; (c) Magnetic field lines; (d) Concentric circles; Field inside solenoid is uniform (C) (a) Magnetic field exists around magnet; (b) Minimum strength near poles; (c) Magnetic field lines; (d) Equidistant parallel lines; Field inside solenoid is uniform (D) (a) Magnetic field exists around magnet; (b) Maximum strength near poles; (c) Magnetic field lines; (d) Equidistant parallel lines; Field inside solenoid is non-uniform
ANSWER: (A) (a) Magnetic field exists around magnet; (b) Maximum strength near poles; (c) Magnetic field lines; (d) Equidistant parallel lines; Field inside solenoid is uniform

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CHAPTER: 13 - Our Environment (Biology)
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YEAR: 2024
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which one of the following is not a natural ecosystem?
OPTIONS: (A) Pond ecosystem (B) Grassland ecosystem (C) Forest ecosystem (D) Cropland ecosystem
ANSWER: (D) Cropland ecosystem

YEAR: 2024
CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A food chain will be more advantageous in terms of energy if it has:
OPTIONS: (A) 2 trophic levels (B) 3 trophic levels (C) 4 trophic levels (D) 5 trophic levels
ANSWER: (A) 2 trophic levels

YEAR: 2024
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Consider the following statements about ozone: (a) Ozone is poisonous gas. (b) Ozone shields the earth's surface from the infrared radiation from the sun. (c) Ozone is a product of UV radiations acting on oxygen molecule. (d) At the lower level of the earth's atmosphere, ozone performs most essential function. The correct statements are:
OPTIONS: (A) (a) and (b) (B) (a) and (c) (C) (b) and (c) (D) (b) and (d)
ANSWER: (B) (a) and (c)

YEAR: 2024
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Some wastes are given below: (i) Garden waste (ii) Ball point pen refills (iii) Empty medicine bottles made of glass (iv) Peels of fruits and vegetables (v) Old cotton shirt. The non-biodegradable wastes among these are:
OPTIONS: (A) (i) and (ii) (B) (ii) and (iii) (C) (i), (iv) and (v) (D) (i), (iii) and (iv)
ANSWER: (B) (ii) and (iii)

YEAR: 2024
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a biodegradable substance?

OPTIONS: (A) Plastic (B) Glass (C) Paper (D) Aluminium can
ANSWER: (C) Paper

YEAR: 2024
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The process by which harmful chemicals accumulate in the bodies of organisms at higher trophic levels is called:
OPTIONS: (A) Bioaccumulation (B) Biomagnification (C) Biodegradation (D) Bioremediation
ANSWER: (B) Biomagnification

YEAR: 2024
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The source of ozone in the upper atmosphere is:
OPTIONS: (A) Industrial activities (B) Combustion of fossil fuels (C) Action of UV radiation on oxygen (D) Volcanic eruptions
ANSWER: (C) Action of UV radiation on oxygen

YEAR: 2024
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Differentiate between food chain and food web. In a food chain consisting of deer, grass and tiger, if the population of deer decreases, what will happen to the population of organisms belonging to the first and third trophic levels?
OPTIONS: (A) Food chain - series of organisms; Food web - network of interconnected food chains; Grass population increases; Tiger population decreases (B) Food chain - network of interconnected food chains; Food web - series of organisms; Grass population increases; Tiger population decreases (C) Food chain - series of organisms; Food web - network of interconnected food chains; Grass population decreases; Tiger

population increases (D) Food chain - series of organisms; Food web - network of interconnected food chains; Grass population increases; Tiger population increases
ANSWER: (A) Food chain - series of organisms; Food web - network of interconnected food chains; Grass population increases; Tiger population decreases

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YEAR: 2024
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Use of pesticides to protect our crops affect organisms at various trophic levels especially human beings. Name the phenomenon involved and explain how does it happen.
OPTIONS: (A) Biological magnification/Biomagnification; Pesticides washed into soil and water; absorbed by plants; non-biodegradable; accumulate at each trophic level; maximum in humans at top level (B) Bioaccumulation; Pesticides washed into soil and water; absorbed by plants; biodegradable; accumulate at each trophic level; maximum in humans at top level (C) Biological magnification/Biomagnification; Pesticides washed into soil and water; absorbed by plants; non-biodegradable; accumulate at each trophic level; maximum in plants at first level (D) Biological magnification/Biomagnification; Pesticides washed into soil and water; absorbed by plants; non-biodegradable; accumulate at each trophic level; minimum in humans at top level
ANSWER: (A) Biological magnification/Biomagnification; Pesticides washed into soil and water; absorbed by plants; non-biodegradable; accumulate at each trophic level; maximum in humans at top level

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